

OLAP CUBE CONSTRUCTION AND ANALYSIS REPORT

Sales Data Analytics Project

Based on the provided Python / Google Colab notebook

1. Introduction

This report presents the OLAP (Online Analytical Processing) cube component of the sales data analytics project. The notebook first establishes a Star Schema containing fact_sales, dim_customer, dim_product and dim_date. The OLAP stage then combines these tables into an analytical dataset and uses pandas.pivot_table to simulate a multidimensional OLAP cube.

2. Purpose of the OLAP Cube

The purpose of the cube is to support multidimensional analysis of sales performance. The notebook describes the cube as a mechanism for aggregating and exploring sales data from different business perspectives, allowing measures to be examined across time and product dimensions, with customer attributes also available in the analytical dataset.

3. Data Model Used

The OLAP stage is based on the Star Schema created in the earlier ETL phases. The fact table is fact_sales, while the main dimensions are customer, product and date.

Component	Role	Relevant Attributes
fact_sales	Central fact table	order_id, quantity, unit_price, sales_amount, total_price
dim_date	Time dimension	year, month_name, quarter, day
dim_product	Product dimension	name, brand, category
dim_customer	Customer dimension	first_name, last_name, city, country

4. OLAP Cube Dimensions

The notebook identifies the following analytical dimensions:

- Date: year, month_name, day and quarter.
- Product: category and brand.
- Customer: city and country.

5. Measures Used in the Cube

The cube definition in the notebook uses the following measures and aggregation functions:

Measure	Aggregation	Business Meaning
sales_amount	SUM	Total sales value
quantity	SUM	Total quantity sold
unit_price	MEAN	Average unit price
order_id	NUNIQUE (described)	Number of unique orders

6. Cube Construction Method

The notebook creates an OLAP-ready denormalized dataset by joining fact_sales with dim_customer and dim_product, followed by a join with dim_date. It then constructs the cube with pandas.pivot_table.

The pivot table uses year, month_name and quarter as row/index dimensions, while category and brand are used as column dimensions. The measures sales_amount, quantity and unit_price are aggregated using SUM, SUM and MEAN respectively.

7. OLAP Operations Demonstrated

Slice: Selects a specific year (2024) and calculates the total sales for that year.

Dice: Selects Electronics products within 2024 to form a more focused sub-cube.

Roll-up: Aggregates the cube by year to provide a higher-level sales view.

Drill-down: Examines monthly sales for a specified product category and brand in 2024.

Category quantity slice: Aggregates total quantity sold by product category across the available years.

8. Interpretation of the Analytical Approach

The cube design supports analysis at multiple levels of granularity. Time can be viewed from year/month/quarter perspectives, while product performance can be examined by category and brand. Customer location attributes are retained in the analytical dataset and can be used for further geographical analysis.

This structure is suitable for business intelligence tasks such as comparing sales between years, identifying category-level demand, examining brand performance, and moving between summary and detailed views.

9. Data Quality Context

Earlier ETL validation in the notebook reports 1,000 transformed sales rows with no rejected rows and no duplicate order_id values. It also reports 638 unique customer IDs and 614 unique product IDs. At the transformation stage, unit_price, sales_amount and total_price contain 1,000 missing values each; the notebook explicitly treats these as expected because price information is intended to be enriched from the product data during integration.

10. Important Implementation Notes

The provided notebook contains the OLAP cube construction code and the intended cube operations, but the cube-building cells themselves do not contain executed output in the provided notebook. Therefore, this report documents the cube design and analytical procedure rather than inventing numerical cube results.

There is also a naming inconsistency in the notebook around the date dimension: one earlier version uses `date_id` as an index, while a later date-dimension section creates `date_key`. The OLAP merge code expects `date_id`. This should be made consistent before executing the complete pipeline end-to-end.

The notebook describes Order Count as a unique count of `order_id`, but the `pivot_table` measure dictionary shown in the cube construction uses `sales_amount`, `quantity` and `unit_price`. If Order Count is required in the actual cube, it should be explicitly added as a measure.

11. Conclusion

The OLAP component extends the Star Schema into a multidimensional analytical structure. Using `pandas.pivot_table`, the project provides a practical simulation of an OLAP cube with time and product dimensions and sales-oriented measures. Slice, dice, roll-up and drill-down operations demonstrate how the same sales data can be viewed at different levels of detail.

For a final production-quality implementation, the date-key naming should be standardized, Order Count should be explicitly implemented if required, and the cube cells should be executed after all ETL and integration stages have completed successfully.

12. Source Basis

This report is based on the uploaded notebook, particularly its Phase 15 OLAP Cube Construction and Analysis section and the preceding ETL/data-quality stages.